

Topology Ph.D. Exam

September 2002

Work the following problems and show all work. Support all statements to the best of your ability. Work each problem on a separate sheet of paper

1. Use the Mayer-Vietoris sequence for reduced singular homology to compute $H_*(S^n)$.
2. Show that there is no retraction of the Mobius band onto its boundary.
3. Does there exist a covering space of the figure eight with nontrivial abelian fundamental group?
4. Show that the 2-dimensional sphere with three points deleted cannot be a topological group.
5. Let M be a 4-dimensional closed simply connected manifold. Show that every continuous map $f : M \rightarrow M$ which is homotopic to the identity has a fixed point.

Answer the following with complete definitions or statements or short proofs.

6. State the Seifert-van Kampen Theorem.
7. State the Lefschetz Fixed Point Theorem.
8. State the Jordan Curve Theorem.
9. State the Arzela-Ascoli Theorem.
10. Let X be a finite complex. Find $\chi(X \times S^n)$.
11. Let $S_r(x, y)$ denote a circle of radius r centered at $(x, y) \in \mathbf{R}^2$. Let ΣR be the suspension over Hawaiian ring $R = \bigcup_{n=1}^{\infty} S_{\frac{1}{n}}(\frac{1}{n}, 0)$. Is ΣR

- (a) locally path connected?
- (b) locally simply connected?
- (c) semi-locally simply connected?

- 12. State the Poincare Duality Theorem.
- 13. Define retraction and deformation retraction.
- 14. State the Universal Coefficient formula for homology.
- 15. Let X be a complete metric space such that no point is isolated. Can X be countable?